

Mohammed Siddiq

AI ML Engineer • MLOps & NLP • Enterprise Systems

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PROFESSIONAL SUMMARY

AI/ML Engineer with a B.Tech in AI & Machine Learning and AWS Cloud Practitioner certification, with experience supporting enterprise systems and building production-focused AI/ML applications. Skilled in Python, NLP, Generative AI, LLMs, RAG, LangGraph, and MLOps, with hands-on experience deploying containerized applications on AWS and integrating live data and external APIs. Built end-to-end AI and data solutions spanning model development, backend APIs, cloud deployment, and system optimization. Seeking AI/ML Engineer roles focused on building reliable, production-ready intelligent systems.

EXPERIENCE

Technical Operations Engineer | Tech Mahindra

Aug 2025 – Present • Hyderabad, India

US-Based Enterprise Client Engagement

- Triaged ~10–12 daily incidents across 2,000+ retail sites by querying structured databases with multi-table SQL joins and aggregate functions, analysing data from multiple sources to pinpoint root causes and dispatch repair services, directly reducing Time to Resolve (TTR) and improving SLA compliance.
- Coordinated with field technicians and lead specialists on escalated, complex issues to drive timely resolution and minimize site downtime across a 30+ member operations team.
- Analysed structured operational data across disparate sources to identify recurring failure patterns and systemic issues, enabling proactive maintenance interventions and reducing repeat incident frequency.
- Maintained accurate technical documentation, incident records, configuration changes, and resolution details to ensure traceability and consistent operational support.

Machine Learning Intern Ignitus

Feb 2024 – Jul 2024 • Remote

- Developed predictive modeling and NLP solutions using Python, Pandas, and Scikit-learn, applying feature engineering to improve model performance.
- Engineered end-to-end data pipelines and ETL workflows to clean, transform, and preprocess structured and unstructured data for machine learning applications.
- Optimized ML inference and data processing pipelines through code refactoring, improving execution efficiency and application performance.
- Collaborated in an Agile development environment using Git, feature branches, pull requests, and peer reviews to build and integrate 3+ reusable ML components for testing and deployment.

Data Analyst & Web Developer (Contract) | Excel Placement Services **India**

Oct 2024 – Mar 2025 • Hyderabad,

- Analyzed recruitment data using SQL, Excel, and Power BI, building KPI dashboards for Time-to-Hire, Submission-to-Interview, and recruitment pipeline performance.
- Automated reporting workflows using Excel, streamlining recruiter reporting processes and improving operational productivity.
- Designed, developed, and deployed a responsive full-stack recruitment website, managing the complete project lifecycle from requirements gathering and UI development to production deployment, contributing to a 40% increase in client acquisition inquiries.
- Led the end-to-end website migration by configuring GoDaddy hosting, DNS records, domain setup, backend components, and production deployment, while optimizing application latency from 1–1.5 seconds to under 500 ms with minimal downtime.

PROJECTS

Masar — Dubai Congestion Prediction & Commute Recommendation System

[Masar](#)

- Engineered and deployed a live-data AI mobility platform using **Python, LangGraph, XGBoost, NetworkX, Docker, Nginx, AWS EC2, DuckDNS, and Let's Encrypt**, exposing the application through a production HTTPS endpoint.
- Built a stateful **LangGraph** workflow combining congestion prediction, confidence-based routing, LLM-assisted context reasoning, graph-based route optimization, and bilingual English/Arabic recommendation synthesis.
- Integrated **TomTom Traffic API, OpenWeatherMap API, Supabase (PostgreSQL), and Groq LLM APIs** to ingest runtime traffic/weather signals, transform live inputs into model features, and generate context-aware commute recommendations.
- Containerized the application with **Docker** and deployed it on **AWS EC2 (t3.micro, Ubuntu 26.04)** behind an **Nginx reverse proxy**, configuring **DuckDNS DNS, HTTP→HTTPS enforcement, and automated Let's Encrypt certificate renewal with Certbot**.
- Implemented production API endpoints for **health checks, live traffic retrieval, location queries, and recommendation generation**, with environment-based configuration and redacted external API reporting for operational debugging.

Tech Stack: Python • LangGraph • XGBoost • NetworkX • Groq API • TomTom Traffic API • OpenWeatherMap API • Supabase • Docker • AWS EC2 • Nginx • DuckDNS • Let's Encrypt

RAG-Powered Multi-Document QA System

[RAG](#)

- Engineered an end-to-end Retrieval-Augmented Generation pipeline with multi-format document ingestion (PDF, DOCX, TXT), semantic vector storage via ChromaDB, and advanced retrieval strategies (MMR + MultiQueryRetriever), delivering cited, cross-document synthesised answers at scale using LLaMA 3.3 70B via Groq API with full source attribution and strategy-switching CLI.

Tech Stack: Python • LangChain • ChromaDB • HuggingFace Sentence Transformers • Groq API • LLaMA 3.3 70B

TECHNICAL SKILLS

Languages: Python, SQL

ML / AI: NLP, Deep Learning, Generative AI, LLMs, RAG, LangGraph, XGBoost, Vector Embeddings, Prompt Engineering, Transfer Learning

Frameworks & Tools: PyTorch, Keras, Hugging Face, LangChain, Pandas, NumPy, Scikit-learn, Flask, NetworkX

MLOps & Cloud: Docker, Docker Compose, GitHub Actions, AWS (EC2, S3, Lambda), REST APIs, Git

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

2020 – 2024

Siddhartha Institute of Technology and Science, JNTUH

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Database Management Systems, Computer Networks

CERTIFICATIONS

- **AWS** Cloud Practitioner Essentials – AWS Training & Certification (Jul 2026)
- Machine Learning Training – Internshala
- Deep Learning with TensorFlow – IBM
- Machine Learning with Python – freeCodeCamp
- Responsive Web Design – freeCodeCamp